



Debre Berhan University

College of Computing Department of Computer Science Internship Report

Project Title: DBU Internship Management System

Hosting Company: ZERGAW CLOUD

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ZERGAW CLOUD, ICT PARK ADDIS ABABA

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Part One:

1. Declaration

I hereby declare that this internship project report titled "DBU Internship Management System" is my original work, carried out during my internship period.

To the best of my knowledge, this report contains no material previously published or written by another person. The codebase for the "DBU Internship Management System" was developed/refined as part of the internship requirements to demonstrate practical application of web development technologies including React, Node.js, and MongoDB.

2. Acknowledgements

I wish to express my sincere gratitude to all those who supported me during my internship at ZERGAW CLOUD. I am especially grateful to my company supervisor, the Website Development team, and all my colleagues who kindly shared their time, knowledge and expertise with me. I also thank my university supervisor for their guidance. Finally, I extend thanks to my family and friends for their continuous encouragement.

3. Executive Summary

This report presents my two month internship experience at ZERGAW CLOUD in Addis Ababa, a leading local cloud service provider in Ethiopia, founded in 2019.

The company offers cloud infrastructure, cloud software, cloud security, broadband internet and demand based ICT projects.

My work was in the Website Development section. I applied tools and techniques such as responsive front end development.

During the internship I improved my practical skills (web development), upgraded my theoretical knowledge, and gained insight into entrepreneurship, work ethics and communication.

Part Two: Company Background

1. Short History

- It was originally founded in **May 2019** under the name ZERGAW ISP.
- First internet provisioning began in September 2019.
- In March 2021 they opened their first cloud infrastructure site at Arada, Addis Ababa.
- In October 2022 they opened their first branch office at Legehâr. In November 2022 they entered into a strategic partnership with Ethio Telecom under the brand "ETZ" (Ethio Telecom Zergaw).
- In February 2023 the company converted to a share company (new corporate structure) and began operating under the trade name ZERGAW CLOUD.
- By June 2023 they indicated having served "500+ enterprise clients".

Thus, in a relatively short span (2019 to 2023) ZERGAW CLOUD has grown from an ISP to a full cloud services provider, establishing infrastructure, offices, and partnerships, aiming to address local needs in Ethiopia's ICT & cloud market.

2. Main Products, Services and Technologies

2.1 Products & Services

- **Cloud Infrastructure** : Highly available and scalable cloud infrastructure that provides your business with the computing and storage resources you need to run your software systems in a cost and time efficient way.
- **Cloud Software** : A suite of software solutions that streamline your business operations, enhance efficiency, improve collaboration, and drive growth.
- **Cloud Security** : Contemporary security solutions to safeguard your business from any attack.
- **Broadband Internet**: For your Broadband Internet and VPN need at your premises to safely access internet services and your cloud infrastructure with quality speed.
- **Demand based ICT Projects**: We are tuned to tune our cloud, internet and other strategically aligned ICT services to your specific demand.

2.2 Technologies

- Hosting infrastructure in "reliable and secured locations of Ethiopia" connected to the national backbone network.
- Bare metal server hardware with remote management interfaces (e.g., DELL iDRAC or HPE iLO). Scalable cloud servers (with SSD or HDD storage, dedicated resources) delivered over Internet or VPN.
- SaaS modules integrated with local payments (e.g., Telebirr), SMS VAS, etc. For example their ERP offering mentions integration with Telebirr and local electronic payments.

3. Main Customers / End Users

- they have served **over 500 businesses** across sectors such as banking, microfinance, hospitality, entertainment, and government institutions.
- clients such as **Wallet Microfinance Institution S.C., Amhara Bank, TSEDEY BANK S.C**

Thus:

- The **end users** include medium to large enterprise organizations (banks, microfinance institutions, government agencies) which have need for reliable infrastructure, software systems, and cloud services.
- They also serve smaller companies and possibly startups
- customers requiring connectivity and cloud hosting are also part of the customer base. Overall, the company targets organizations that need cloud hosting, ERP and digital transformation services in the Ethiopian context.

4. Organizational Structure

- **Founders:** Co Founders are Dr. Beneyam Berehanu (PhD) and Dr. Ephrem Teshale (PhD) as respectively CBDO and CTO.
- **CEO:** Addisu Kassaw
- **Other senior managers:** Miraf Kena (Customer & Partnership Manager), Eyerusalem Michael (Operations Manager), Nardos Endale (Finance Manager), Yared Mengistu (Software Engineer Team Leader)

a simplified organizational structure:

- **Board or Shareholders** – as it converted to a share company in 2023.
- **Executive Management** – CEO, CTO, CBDO.
- **Departments** – Customer & Partnerships; Operations; Finance; Technical/Engineering (software, infrastructure); Sales & Marketing; Support.
- **Subsidiary / Office Locations** – Headquarters in ICT Park (Addis Ababa), branch office at Legehâr, data centre sites in Arada & Legehâr.

5. Workflows

5.1 Service Onboarding

- A customer (enterprise) approaches ZERGAW with a requirement (e.g., cloud server, ERP, storage).
- Technical team assesses customer needs – computing, storage, connectivity, security. The infrastructure is offered via scalable cloud servers or bare metal as per requirement.
- Customer signs service agreement, chooses pay as you go or subscription plan (they emphasise local payment options, local currency).
- Deployment: provisioning cloud servers, storage or software modules. The website indicates rapid deployment ("Short deployment time") and high availability.
- Handover and training/support to customer. Post deployment, customer accesses their server/solution.

5.2 Service Operation & Maintenance

- Infrastructure monitoring: The cloud infrastructure operates from data centres connected to the national backbone network. Uptime is measured (99.95% availability).
- Customer support: 24/7 support is emphasised (for example cloud services page mentions 24/7 support).
- Scaling / modifications: Customers can scale computing/storage resources up or down as needed.
- Security and compliance: Security services and shared responsibility model emphasised in cloud infrastructure descriptions.
- Billing & payments: Services are billed , likely via local currency, with customer support and finance teams handling the process.

5.3 Custom Projects

- For demand based ICT projects: customer requirement gathering 'n' design 'n' development (software & infrastructure) 'n' deployment 'n' maintenance.

5.4 Internal Development & QA

- In house software/web development team: they have web designers, developers, copywriters etc.
- Quality assurance and product engineering: Part of their value proposition emphasizes product improvements through testing, refinement.

Part Three: Internship Experience

1. Department / Section Worked In & Why

During my **two month internship**, I was assigned to the Website Development section of ZERGAW CLOUD.

I selected this section because I have a keen interest in web development and wished to gain practical experience building real world web projects

2. Work Tasks Executed

- Performed testing of websites for cross device compatibility (desktop, tablet, mobile) and verified responsiveness.
- Participated in client handover workflows: ensuring domain, SSL certificate installation, DNS configuration.

3. Systems Development, Tools & Techniques Used

- A CMS (content management system)
- Responsive front end web technologies (HTML, CSS, JavaScript) to ensure websites work on every device ("Works on every device and platform").

4. Major Challenges and Problems Faced

- **Time constraints:** As an intern, I had to adapt quickly and deliver components under tight deadlines.
- **Cross device compatibility:** Ensuring fully responsive layout across a wide variety of devices (older browsers/devices common in the local market) required deeper testing than originally planned.

5. Measures & Proposed Solutions to Overcome Challenges

- For time constraints: I adopted a **mini iteration approach** – breaking the website project into smaller deliverables (home page, responsive layout, CMS integration, content upload) so that parts could go live earlier and adjustments done afterward.
- For compatibility testing: I increased use of **device emulators** and actual device testing (older Android phones, tablets), and created a compatibility checklist for the locally common browsers/devices.

6. Improvement in Practical Skills

- I became comfortable with building responsive websites from scratch.
- I developed skills in performance optimization, cross browser/device testing, SSL installation, domain/DNS configuration.

7. Upgrading Theoretical Knowledge

- I refreshed my understanding of responsive web design theory.
- I learnt more about service level agreements (SLA) and how deployment timelines, uptime, and scalability matter for website services tied to cloud infrastructure. For example, the company claims 99.95% availability for its infrastructure.

8. Team Playing Skill Improvements

- I learned to manage dependencies (e.g., design assets arriving before front end coding, hosting team provisioning before deployment) and communicate status updates proactively.

9. Leadership Skill Improvements

- I observed the company's management and learnt how a professional cloud/ICT service company sets direction, prioritizes value (affordability, localization, scalability) and leads the team.

10. Understanding Work Ethics, Company Psychology & Related Issues

- The company emphasizes **affordability, quality, localization, security, scalability** as core values.
- I observed how delivering quality.
- I learnt the importance of **system uptime, deployment speed and customer satisfaction**, which are core for a cloud/hosting business operating in the local market.
- I also saw how a local cloud company in Ethiopia positions itself: focusing on local payments, local hosting, and matching local market needs (which changes the psychology of product delivery compared to a global provider).

11. Entrepreneurship Skills

- I saw how ZERGAW CLOUD built a website service product with pricing tiers (Basic, Standard, Premium) for different customer segments.
- I learned about scaling a service: how the website service can be hosted locally, delivered affordably, and scaled for more pages or advanced features — this gives ideas of how one might launch or expand a service in such a market.
- I also appreciated the importance of aligning technology services to local market constraints (e.g., payment in local currency, local hosting) which is vital for entrepreneurship in Ethiopia.

12. Recommendations & Conclusion

12.1 Recommendations:

- Formalize a **mentorship/training program** for interns, orienting them quickly into the company's website service process flow, tools and expectations.
- Introduce more **automation** in the hosting/deployment workflow (e.g., scripts to set up hosting, SSL, domain configuration) to reduce time and manual errors.
- Strengthen **device testing environments** by maintaining a pool of locally common mobile devices/browsers so interns can test on real hardware rather than only emulators.
- Create a **knowledge base** of previous website projects (screenshots, challenges, lessons learnt) so future interns can leverage past experience.

12.2 Conclusion:

I entered the internship with theoretical knowledge of web development and a desire to apply it in a real business context. I have improved my practical web development capabilities, expanded my theoretical understanding of cloud hosted services.

PART FOUR:

PROJECT WORK (DBU Internship Management System)

1. Summary of the Project

The project developed during the internship is titled **DBU Internship Management System (DBU-IMS)**. The purpose of this system is to digitize and automate the internship management process used by Debre Birhan University.

Currently, most internship coordination processes in the university are handled using **manual methods such as Excel spreadsheets, and Telegram groups**. These approaches lead to several issues such as poor tracking of student progress, difficulty managing reports, lack of centralized data storage, and inefficient communication between students, advisors, and department administrators.

The DBU-IMS system is designed to solve these issues by providing a **centralized web-based platform** where students, advisors, department deans, and administrators can manage internship activities efficiently.

The system provides the following major functionalities:

- Internship application management
- Advisor assignment and supervision
- Weekly and final report submission
- Internship logbook management
- Internship evaluation and grading
- Communication between users
- Internship progress monitoring and analytics

The system is developed using the **MERN Stack technology**, which includes **MongoDB, Express.js, React.js, and Node.js**, enabling the creation of a modern, scalable, and secure web application.

2. Problem Statement and Justification

2.1 Existing System

The current internship management process at many universities, including Debre Birhan University, is largely **manual and fragmented**. Communication between students, advisors, and administrators often occurs through **Telegram groups, or physical documents**.

Some of the major problems in the existing system include:

- Lack of a centralized platform to manage internship activities
- Difficulty tracking student internship progress
- Inefficient submission and review of reports
- Poor record keeping of student internship data
- Limited communication structure between stakeholders
- Risk of data loss when using spreadsheets or messaging applications
- Difficulty generating reports and monitoring internship performance

Because of these limitations, both students and academic staff spend significant time managing administrative tasks instead of focusing on the learning objectives of the internship program.

2.2 Justification for the Proposed System

The DBU Internship Management System addresses these problems by providing a **centralized digital platform** that automates internship management processes.

The proposed system improves the current situation by:

- Centralizing internship data in one system
- Automating internship application approval workflows
- Providing secure report submission and storage
- Allowing advisors to review and evaluate student progress easily
- Enabling department administrators to monitor internship activities
- Improving communication between stakeholders
- Generating reports and statistics for decision-making

Therefore, implementing this system significantly improves the efficiency, transparency, and reliability of the internship management process.

3. Objective of the Project

3.1 General Objective

To design and develop a **web-based Internship Management System** that supports the efficient management of internship activities at Debre Birhan University.

3.2 Specific Objectives

- To create a centralized platform for managing internship applications
- To enable students to submit internship reports digitally
- To allow advisors to review and evaluate student reports
- To automate advisor assignment and internship approval processes
- To provide dashboards for monitoring internship progress
- To improve communication between students, advisors, and administrators
- To generate internship performance reports and analytics

4. Methodology

The development of the DBU-IMS system followed a structured methodology involving requirement analysis, system design, implementation, and testing.

4.1 Fact-Finding Techniques

Several fact-finding techniques were used to understand the internship management process and identify system requirements.

These techniques include:

Observation

The internship coordination process within the organization was observed to understand how internship applications, reports, and evaluations are currently handled.

Document Analysis

Existing internship forms, evaluation sheets, and guidelines were reviewed to understand the current workflow and data requirements.

4.2 Tools, Languages, and Frameworks Used

The project was developed using modern web development technologies.

Frontend Technologies

- **React.js** – used to build the user interface
- **Tailwind CSS** – used for responsive and modern UI design
- **Vite** – used as the frontend build tool

Backend Technologies

- **Node.js** – used to run the server environment
- **Express.js** – used to build RESTful APIs

Database

- **MongoDB Atlas** – used as the cloud database for storing system data

Authentication

- **JWT (JSON Web Token)** – used for secure authentication and authorization

Other Tools

- **Git & GitHub** – version control and project repository management
- **Cloudinary** – file storage for report uploads
- **Postman** – testing API endpoints

4.3 System Design

The system follows a **Role-Based Access Control (RBAC)** architecture. Each user role has specific permissions and functionalities.

The four main roles in the system include:

- Student
- Advisor
- Department Dean
- College Head/Admin

The system architecture consists of three main layers:

Presentation Layer – React frontend user interface

Application Layer – Node.js and Express backend services

Data Layer – MongoDB Atlas database

4.4 User Authentication and Account Management

In the DBU Internship Management System, usernames are automatically generated when the College Head/Admin creates student accounts. The username follows the format "DBU + seven digits". After account creation, the system provides the usernames through the admin dashboard.

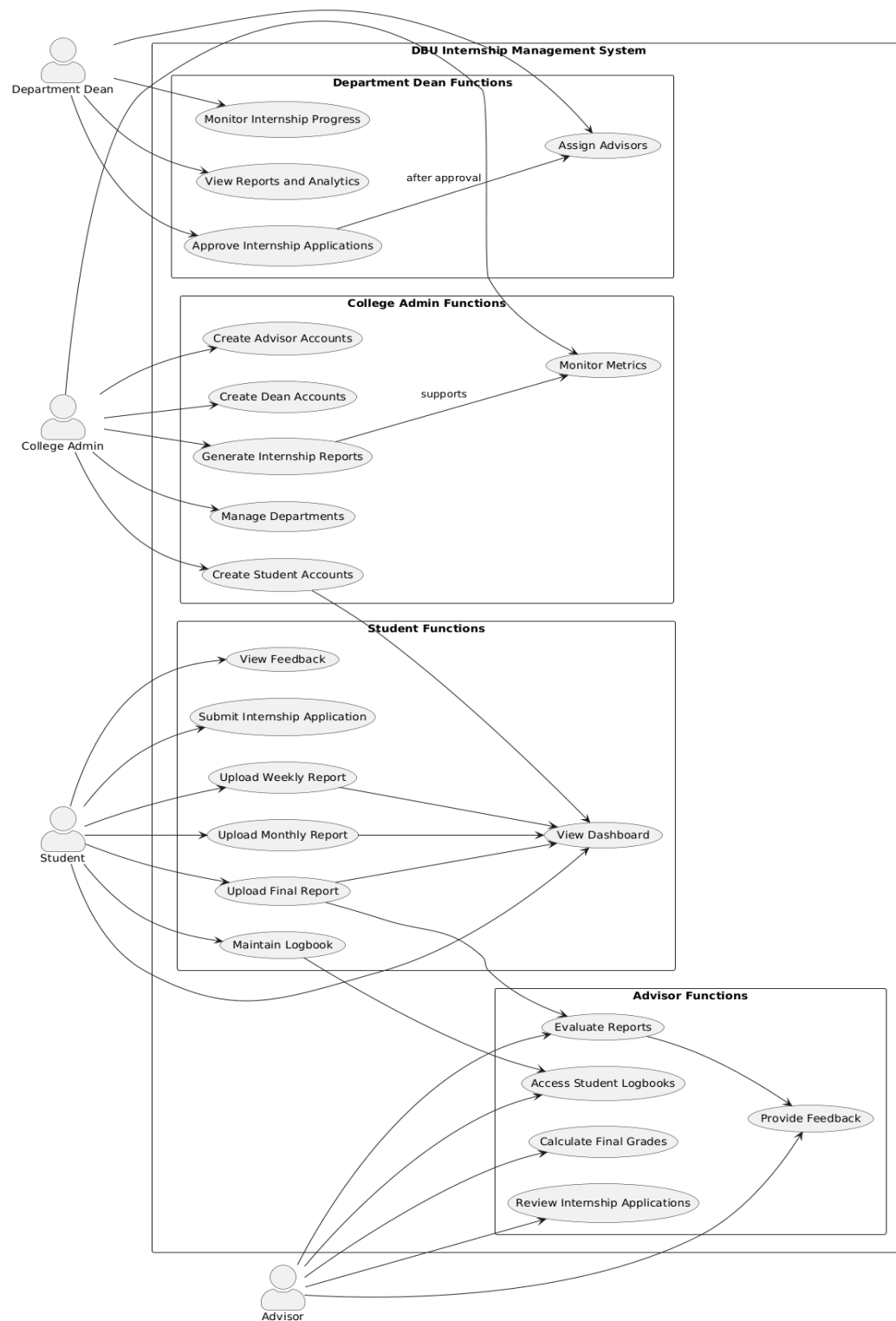
The administrator can print a list containing student names, student IDs, and usernames. This list is shared with students so that they can activate their accounts. During account activation, students enter their username, student ID, and password to create their system account.

4.5 Authentication System

The DBU Internship Management System implements a role-based authentication mechanism. Each user receives a unique username generated by the system. Users activate their accounts during the first login by providing their username, identification information, and password.

After activation, users access the system using their username and password. The system supports multiple roles including student, advisor, department dean, and college administrator. Each role has different permissions and access levels.

5. Usecase diagram of the system



6. Results and Discussion

The developed DBU-IMS system successfully provides a platform that automates and simplifies internship management processes.

The system allows students to submit internship applications, upload reports, and maintain logbook records. Advisors can review student submissions, provide feedback, and evaluate internship performance. Department administrators can monitor internship activities and assign advisors efficiently.

The system also includes dashboards that provide insights into internship progress and student performance.

Compared to the traditional manual process, the developed system offers several advantages:

- Faster internship application processing
- Improved record management
- Secure storage of internship documents
- Better communication between stakeholders
- Real-time monitoring of internship activities

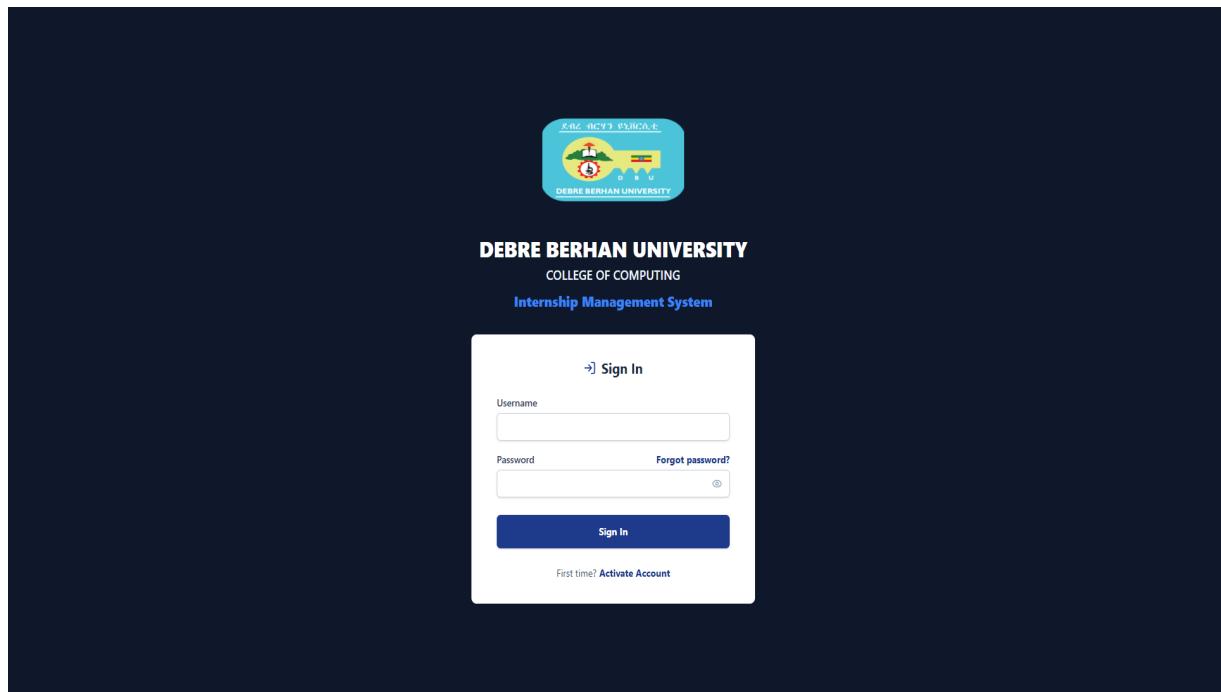
The implementation of this system demonstrates how digital technologies can significantly improve academic administrative processes.

7. Screenshots of the Project

7.1. Login Page

The entry point for all system users, featuring a clean, professional interface aligned with DBU branding.

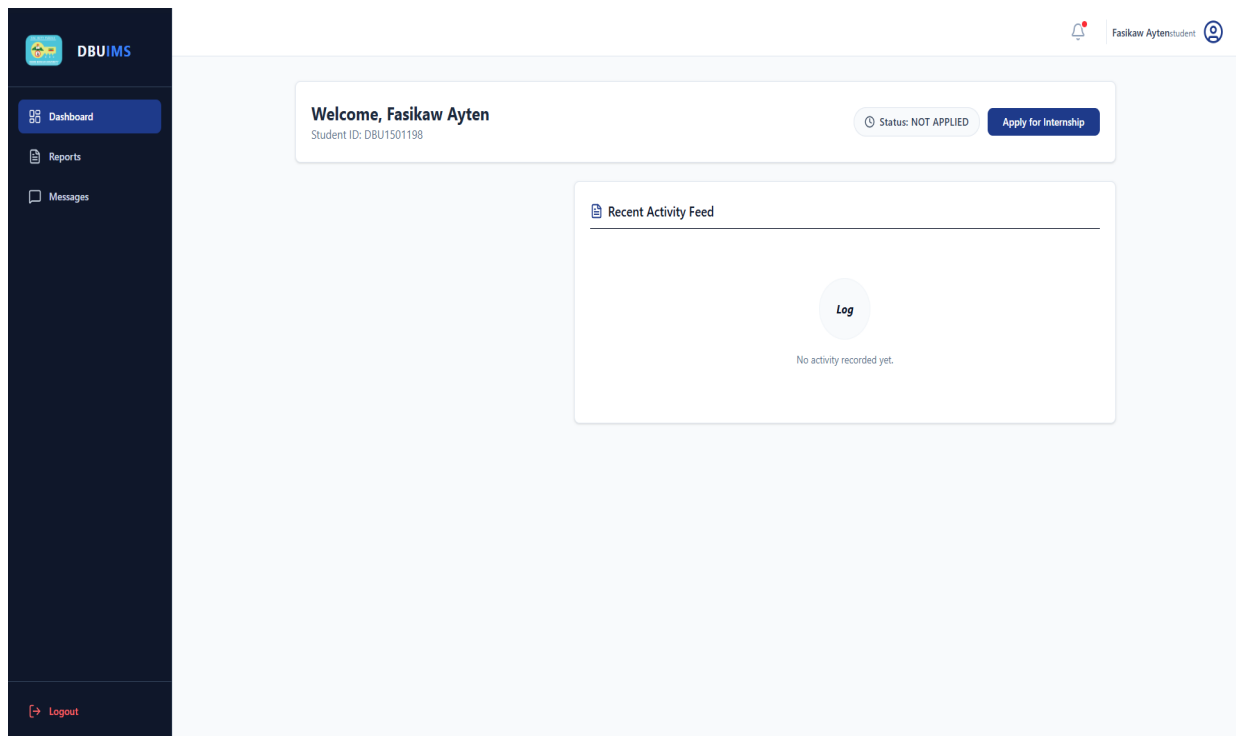
- **Dual Flow:** Supports standard login for activated accounts and a dedicated **Account Activation** flow for first-time student/staff entry.
- **Security:** Implements JWT-based session management and "Forgot Password" functionality that verifies account activation status.
- **Role Detection:** Automatically redirects users to their specific dashboard based on their system role after successful authentication.



7.2. Student Dashboard

A centralized hub for interns to manage their entire internship lifecycle.

- **Status Tracking:** Visual indicators for internship progress (e.g., PENDING, ONGOING, COMPLETED).
- **Profile Settings:** Dedicated interface for students to maintain their phone number and **CBE Account Number** (Commercial Bank of Ethiopia) for stipend processing.
- **Metrics:** Real-time display of total hours worked and reports submitted.



7.3. Internship Application Form

The formal interface where students propose their internship placement.

- **Company Details:** Collection of company name, location, and industry/field.
- **Supervisor Information:** Capture of on-site supervisor contact details for university verification.
- **Workflow:** Upon submission, the application is routed to the Department Dean for approval.

The screenshot displays the DBUIMS (Dakota Baptist University Institutional Management System) interface. On the left is a dark blue sidebar with navigation links: Dashboard, Reports, Messages, and Logout. The main content area has a light blue header with a welcome message for 'Fasikaw Ayten' (Student ID: DBU1501198) and a status indicator 'Status: NOT APPLIED'. Below this is a modal window titled 'Internship Application' with a 'CANCEL' link. The form contains several input fields: Company Name, Location, Field of Work, Company Supervisor Name, Supervisor Phone, Supervisor Email, Start Date (with a date picker), and End Date (with a date picker). A large blue button at the bottom of the modal is labeled 'Submit Internship Details'.

7.4. Report Upload Interface

A robust module for document submission and version control.

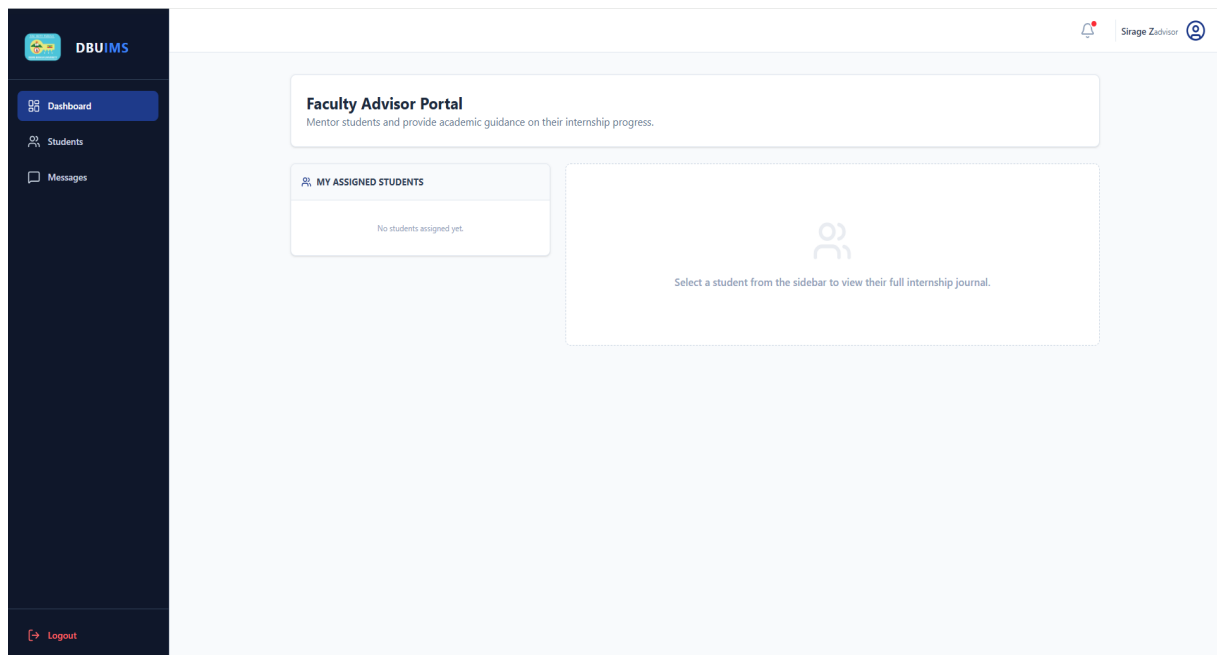
- **Categorization:** Specific sections for **Weekly**, **Monthly**, and **Final** reports.
- **Versioning:** Allows students to upload revisions; the system automatically tracks version numbers and highlights the "Latest" version for advisors.
- **Deadline Awareness:** Displays due dates and includes automated "Late Submission" flagging.

The screenshot displays the 'Submit Report' interface within the DBUIMS system. On the left is a dark blue sidebar with navigation links: 'Dashboard', 'Reports' (highlighted), and 'Messages'. At the bottom of the sidebar is a 'Logout' link. The main content area is light blue and features a central white box titled 'Submit Report' with a cloud upload icon. Below the title, it says 'Upload your PDF/DOCX reports. Version tracing is automatic.' The form includes a 'Report Type' dropdown menu set to 'Weekly Report', an 'Assigned Due Date' field with a date picker icon and placeholder 'mm/dd/yyyy', and a note: 'If submitted past this date, the system flags it automatically.' Below these fields is a dashed box with the text 'Click to select file (PDF/DOCX)' and a large blue 'Upload Report' button. In the top right corner of the page, there is a notification bell icon and a user profile for 'Fasikaw Aytenstudent'.

7.5. Advisor Dashboard

A workstation for university advisors to oversee their assigned students.

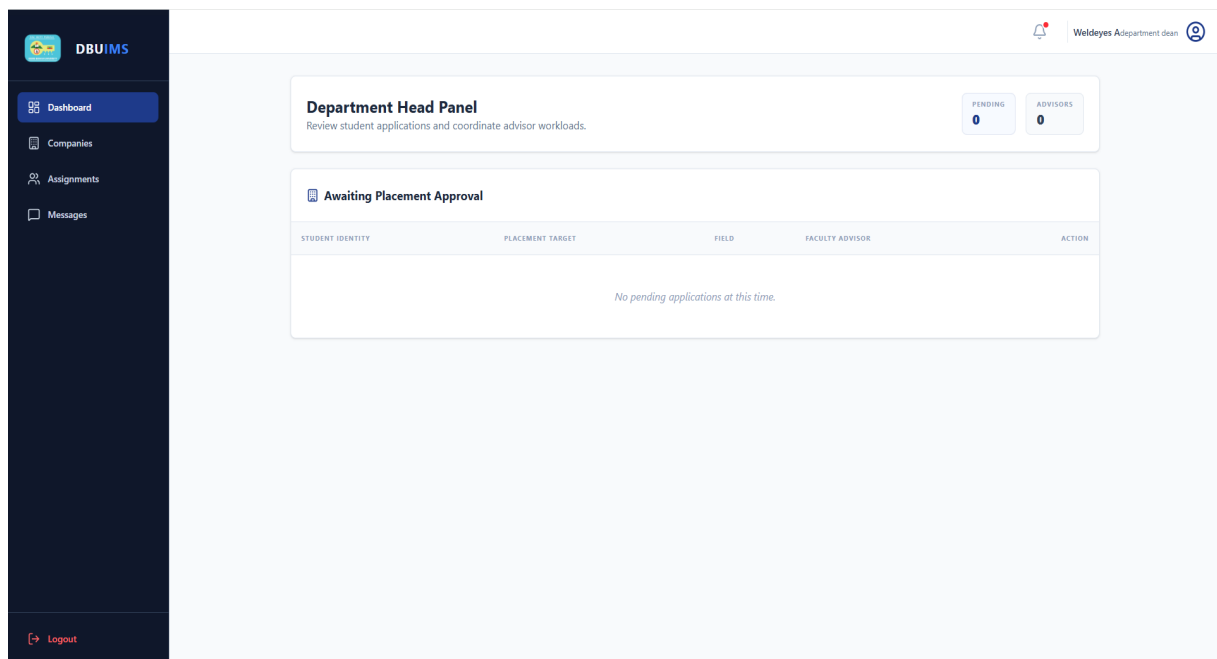
- **Student List:** A high-level view of all assigned interns, their current status, and quick links to their reports.
- **Grading Portal:** An integrated form for finalized evaluations, featuring automatic calculation based on the 4-component weighted system (30/25/25/20).
- **Feedback Loop:** Direct access to student logbooks and reports for real-time mentoring.



7.6. Department Dean Dashboard

A command center for Department Heads to manage internship placements within their department.

- **Placement Approval Queue:** A table of pending student applications with company details and a one-click Approve action.
- **Advisor Assignment:** A built-in dropdown per student to instantly assign a faculty advisor after approval.
- **Live Stats:** Header badges display the total number of pending applications and available advisors at a glance.



7.7. Admin Dashboard

The "Mission Control" for the College Head.

- **System Metrics:** Visual overview of student enrollment, department counts, and staff distribution.
- **Provisioning:** Centralized tool for creating Student, Advisor, and Dean accounts with unique, system-generated usernames (e.g., DBU+7 digits).
- **Department Control:** Creation and management of college departments and their corresponding heads.

The screenshot displays the DBUIMS College Administrative Console. On the left is a dark sidebar with navigation links: Dashboard, Staff, Seed IDs, Messages, and a Logout button at the bottom. The main content area has a header titled "College Administrative Console" with a subtitle: "Monitor university-wide metrics, oversee computing department performances, and securely provision staff and student access." Below the header are two primary forms. The "Provision New Access" form on the left includes fields for Full Name (John Doe), System Role (Student), Academic Email (name@dbu.edu.et (Optional)), Student ID (DBU1234567), CBE Account Number (1000...), and a Department dropdown menu, followed by a "Provision student Access" button. The "Create Department" form on the right includes fields for Department Name (Computer Science) and Department Code (CS), followed by a "Create Department" button. The top right corner shows a notification bell, the user "Dr. Seble Ecollege admin", and a profile icon.

8. Conclusion and Recommendation

The DBU Internship Management System provides a modern solution to the challenges associated with manual internship management processes.

The system successfully demonstrates how a web-based platform can improve efficiency, communication, and data management within academic internship programs.

The use of modern technologies such as React, Node.js, and MongoDB allows the system to be scalable, maintainable, and secure.

It is recommended that the system be further enhanced with additional features such as mobile support, email notifications, and advanced reporting capabilities to further improve its usefulness.

PART FIVE:

GENERAL CONCLUSION AND RECOMMENDATION

General Conclusion

The internship program provides students with valuable opportunities to gain practical experience and apply theoretical knowledge in real-world environments. During the internship period, students are exposed to professional working environments where they develop technical, communication, and teamwork skills.

The development of the DBU Internship Management System as part of the internship project demonstrates the practical application of software engineering principles and modern web development technologies.

The system addresses the limitations of traditional internship management methods by providing a centralized digital platform that supports efficient coordination between students, advisors, and administrators.

Through the development of this project, significant improvements were achieved in understanding system design, full-stack web development, and software project implementation.

Recommendations

Based on the experience gained during the internship and the development of the DBU-IMS system, the following recommendations are proposed:

- Debre Birhan University should consider adopting a digital internship management platform to improve administrative efficiency.
- Future improvements should include mobile application support to allow users to access the system from smartphones.
- Email and notification systems should be integrated to inform users about important internship deadlines and updates.
- Security measures such as two-factor authentication can be implemented to enhance system security.
- Continuous system maintenance and updates should be carried out to ensure long-term reliability.

Implementing these recommendations will help improve the effectiveness and sustainability of internship management processes.